VISHWANATHAN N

+91 9790739680 | vishwanathan247@gmail.com | LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Frontier Engineer Intern focused on enterprise GenAI solution engineering across RAG, agentic AI, and Model Context Protocol (MCP) ecosystems. Experienced in building LLM workflows with LangChain retrieval, MCP tool orchestration, CrewAI/AutoGen agents, guardrails, audit logging, validation, and Gradio/Streamlit interfaces. Able to translate business scenarios into grounded AI architectures, client demos, validation plans, and reusable solution patterns.

TECHNICAL SKILLS

Languages: Python, SQL, HTML, CSS
GenAI Frameworks: LangChain, LangGraph, CrewAI, AutoGen, OpenAI/Azure OpenAI-compatible models
AI Systems: RAG, Agentic AI, MCP/FastMCP, Tool-Using Agents, Prompt Orchestration, Hybrid Retrieval
Engineering: Pydantic, LanceDB, Chroma, RAGAS, SQLite, Pytest, Guardrails, Audit Logging, Docker, Git
Interfaces: Gradio UI, Streamlit

PROJECT EXPERIENCE

Nexus – Unified Financial Services Concierge GitHub

- Designed a financial-services concierge where one free-text customer request flows through Pydantic intake, LangChain + MCP hybrid retrieval, five CrewAI agents, and four scoped MCP servers.
- Implemented architecture for 13 tools, 4 resources, 2 prompts, DB isolation, ownership checks, escalation handling, guardrails, DEBUG JSON tracing, and PII-safe audit logging.
- Built validation coverage with 68 local tests across config loading, server surfaces, pipeline scenarios, hard escalation, conflict handling, offline-server behavior, KYC, and prompt-injection cases.

BankForge – MCP Banking Demo Suite GitHub

- Built an MCP-based banking demo ecosystem with Accounts, Products, and Compliance/Communications servers for enterprise banking walkthroughs.
- Implemented runtime validators and inspection utilities to enumerate and verify MCP tools, resources, prompts, templates, and runtime behavior before demos.
- Contributed communication templates, audit helpers, guardrail utilities, Docker artifacts, and Pytest coverage for controlled customer-interaction workflows.

CatalogueIQ – RAG Product Intelligence Assistant GitHub

- Built a RAG-powered assistant for ShopSmart India to answer product specs, price comparisons, recommendations, return/refund policies, seller guidance, and category knowledge queries.
- Implemented a LangChain retrieval pipeline with LanceDB vector storage, background ingestion, source references, retrieved context display, and optional category/source/price filters.
- Added Gradio UI and RAGAS evaluation workflows to support grounded answer review, faster demos, and measurable retrieval-quality assessment.

CERTIFICATIONS

NPTEL: Database Management Systems
Oracle: Generative AI Certification

EDUCATION

B.Tech in Information Technology Panimalar Engineering College, Chennai
CGPA: 8.8
Class 12: Pachaiyappa's Hr. Sec. School – 91%

KEY STRENGTHS & ACHIEVEMENTS

Research Publication: Published paper on "Open Road Tolling System" at IConiC Conference 2024.
Academic Excellence: Achieved 3rd rank in the department during the 1st semester.